IMO(58^{th}) – 2017

First Day

1. For each integer $a_0 > 1$, define the sequence $a_0, a_1, a_2, ...$ by:

$$a_{n+1} = \begin{cases} \sqrt{a_n} & if \sqrt{a_n} \text{ is an integer} \\ a_n + 3 & otherwise \end{cases}$$, for each $n \geq 0$.

   Determine all values of a_0 for which there is a number A such that $a_n = A$ for infinitely many values of *n*.

2. Let R be the set of real numbers. Determine all functions $f : R \to R$ such that, for all real numbers *x* and *y*, $f(f(x)f(y)) + f(x+y) = f(xy)$.

3. A hunter and an invisible rabbit play a game in the Euclidean plane. The rabbit's starting point A_0, and the hunter's starting point B_0, are the same. After $n-1$ rounds of the game, the rabbit is at point A_{n-1} and the hunter is at point B_{n-1}. In the n^{th} round of the game, three things occur in order.
   (a) The rabbit moves invisibly to a point A_n such that the distance between A_{n-1} and A_n is exactly 1.
   (b) A tracking device reports a point P_n to the hunter. The only guarantee provided by the tracking device to the hunter is that the distance between P_n and A_n is at most 1.
   (c) The hunter moves to a point B_n such that the distance between B_{n-1} and B_n is exactly 1.

Second Day

4. Let R and S be different point on a circle Ω such that RS is not a diameter. Let *l* be the tangent line to Ω at R. Point T is such that S is the midpoint of the line segment RT. Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects *l* at two distinct points. Let A be the common point of Γ and *l* that is closer to R. Line AJ meets Ω again at K. Prove that the line KT is a tangent to Γ.

5. An integer $n \geq 2$ is given. A collection of $N(N+1)$ soccer players, no two of whom are of the same height, stand in a row. Sir Alex wants to remove $N(N-1)$ players from this row leaving a new row of $2N$ players in which the following N conditions hold:
   (1) No one stands between the two tallest players.
   (2) No one stands between the third and fourth tallest players.

   $\vdots$

   (N) No one stands between the two shortest players.

   Show that this is always possible.

6. An ordered pair (x, y) of integers is a primitive point if the greatest common divisor of x and y is 1. Given a finite set S of primitive points, prove that there exist a positive integer n and integers $a_0, a_1, a_2, ..., a_n$ such that, for each (x, y) in S, we have:

$$a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + ... + a_{n-1} x y^{n-1} + a_n y^n = 1$$

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